

Project | Module: Robot project Overview

This semester you will be solving an open-ended design project developed out of a needs assessment of your choice. Your team must use Robotic elements, in that it must use some motors, sensors, servos and have it be programmed. the first part will be finding and proposing that project, then prototyping and presenting your teams design.

✓ Objectives

1. Learn and apply the steps of the engineering design process in proposing and building solutions, working devices, and/or models with consideration of public health, safety, and welfare, as well as global, cultural, social, environmental, and economic factors.
2. Research the scientific principles, technical background required to understand the problem to be solved, and benchmark existing or related products.
3. Apply value-sensitive design principles to determine clients' and users' needs and then identify predetermined specifications, functions, objectives, and constraints of any potential solution by writing a Problem Statement.
4. Generate several alternative solutions that could potentially satisfy the Problem Statement by using various ideation techniques.
5. Decide which solution(s) to pursue by comparing alternative solutions and their probability of success using specifications, function, objectives & constraints from the Problem Statement.
6. Implement the desired solution(s) by building and testing models and/or prototypes utilizing basic hand tools, rapid prototyping, and other technologies, as appropriate.
7. Evaluate the desired solution(s) by assessing feedback, test results, outcomes, and potential safety hazards relative to the Problem Statement and professional standards and codes of engineering conduct.
8. Create and deliver effective individual and team presentations on engineering projects and topics;
9. Clearly document and effectively report engineering projects in interim and final reports that include applicable technical drawings, graphs, and descriptions of design projects.

Project Description

- **Robotics Projects**

Design Project Overview for Cornerstone - Spring 2024

This semester you will be solving an open-ended design project developed out of a needs assessment of your choice. Your team must use Robotic elements, in that it must use some motors, sensors, servos and have it be programmed using programming boards (Redboards or Arduinos if you have access), you are not limited to the number of boards you can use. You will start with your Sparkfun kits. You may use other robotics parts, kits or components to accomplish your design. Your team budget limit is \$100, but cost effectiveness should be part of your goals and you can use as little or as much of this limit as your team sees fit. Your proposal will be reviewed for feasibility and you will be given recommendations on what kind of sensors and components will best fit your needs. Consider having the robot perform a humanitarian function or in some way a service to others. Think about the ethical implications of your design and its functions. Reflect back on the Robotic presentations and look for where the best robotic applications may exist. The following projects are just suggestions and you are encouraged to come up with your own projects and ideas. These must be original and built by your team for most of the device (over 60%), not an addition to an existing robotic device, such as a drone.

For each proposed design, you will need to research the progress made in that area, and what already exists as prototyped, marketed or patented. This does not preclude your design area, but is necessary to guide your own design objectives. Oftentimes there are design ideas and existing devices, so that you will need to adjust your project in order to compensate for that.

At the end of the semester, you will be presenting your robot (or robot prototype design). As you design and build, you will need to plan on how this will present most effectively and develop an environment to demonstrate it as well.

Here are some project ideas:

- **Design for Resilience**

Resilient design is the intentional design of buildings, landscapes, communities, and regions in order to respond to natural and manmade disasters and disturbances—as well as long-term changes resulting from climate change—including sea level rise, increased frequency of heat waves, and regional drought.

- **Robotic Assistive Devices**

Robotic devices are being used more and more to assist people with a wide range of disabilities. Your group must research an area, see what exists and prototype a device.

- **Design for Investigation in Hazardous Area using Robotics**

Design a robot to investigate and operate in a hazardous area such as a nuclear waste site,

an unstable area (landslide, avalanche) or chemical spill.

- **Design for Investigation in Underground situation using Robotics**

Design a robot that will investigate and operate in an underground area such as a mine, cave or deep under the earth.

- **Design for Investigation using Robotics – Your idea**

Design a robot that will autonomously operate and solve a problem of your choice as to the application area, as seen in other ideas.

- **Design for Robotics in Manufacturing**

Design a robot that will autonomously operate in a manufacturing plant or commercial operation.

- **Design for Robotic Underwater work, or Water cleanup work**

Design a robot that will accomplish needed tasks underwater or in an area where water cleanup is required

- **Design for Robotic Construction**

Difficult work, involving heavy lifting and repetition can often be work for robots. There are existing solutions, as you saw in earlier presentations, what other fields are there?

- **Design for Investigation on Mars using Robotics**

Design a platform that will autonomously operate on Mars (or other planet) that might investigate the environment, collect samples and return data to be analyzed or analyzed in real time.

- **Design for Pandemic/Endemic Use**

Design a robot or robotic function to assist in any aspect of the Pandemic. Health Care, testing, vaccinations, distribution of materials, collecting and analyzing data